Project: Worked examples
Title: Rolling bearing life (ISO 281)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The basic and modified rating life of a rolling bearing under a steady load or a duty cycle, at a chosen reliability, against a required life; and the static safety factor against a required value.
Equivalent load: P = X_1·F_r + Y_1·F_a when F_a/F_r ≤ e_lim (X_1 = 1, Y_1 = 0 for a single-row radial ball bearing), else P = X·F_r + Y·F_a (Y = 0 is refused when an axial load acts above e_lim) Duty cycle: P_eq = (Σ P_i^p·U_i / Σ U_i)^(1/p) Basic rating life: L_10 = (C/P)^p million revolutions, p = 3 ball, 10/3 roller Life in hours: L_10h = 10^6·L_10 / (60·n) Reliability factor (ISO 281:2007, derived): a_1 = 0.95·[ln(100/R)/ln(100/90)]^(2/3) + 0.05 Modified life: L_nm = a_1·a_ISO·L_10 Static safety: P_0 = max(X_0·F_r + Y_0·F_a, F_r), s_0 = C_0/P_0

Ref: ISO 281:2007 cl. 6–7 — Dynamic equivalent load and basic rating life
Ref: ISO 281:2007 cl. 9 — Modified rating life
Ref: ISO 76:2006 cl. 6–7 — Static equivalent load and static safety factor
Ref: Shigley's Mechanical Engineering Design 11th ed., Ch. 11 — Rolling-contact bearings

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Bearing type	ball (p = 3)		assumption: Deep groove ball bearing.	any
F_r	Radial load	2.5	kN	assumption: Radial reaction from the shaft calculation.	0 … 2000 kN
F_a	Axial load	0	kN	assumption: Locating bearing elsewhere; no axial load here.	0 … 2000 kN
n	Speed	1500	rpm	spec: Pump datasheet, rated speed	1 … 50000 rpm
C	Basic dynamic load rating	20.3	kN	spec: SKF catalogue, 6206	0.1 … 10000 kN
C_0	Basic static load rating	11.2	kN	spec: SKF catalogue, 6206	0.1 … 10000 kN
e_lim	Load ratio limit e	0.22		spec: Manufacturer factor for single-row deep groove bearings	0 … 5
X_1	Radial factor X_1 for F_a/F_r ≤ e (1 for a single-row radial ball bearing)	1		default	0 … 5

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
Y_1	Axial factor Y_1 for F_a/F_r ≤ e (0 for a single-row radial ball bearing; above 0 for double-row and paired bearings)	0		default	0 … 10
X	Radial factor X for F_a/F_r > e	0.56		spec: Manufacturer factor for single-row deep groove bearings	0 … 5
Y	Axial factor Y for F_a/F_r > e (from the manufacturer; Y = 0 makes the axial load count for nothing and is refused when one acts)	2		spec: Manufacturer factor for single-row deep groove bearings	0 … 10
X_0	Static radial factor X_o	0.6		spec: Manufacturer static factor	0 … 5
Y_0	Static axial factor Y_o	0.5		spec: Manufacturer static factor	0 … 10
R	Reliability	95	%	assumption: Client reliability target.	90 … 99.95 %
a_iso	Life modification factor a_ISO (from the manufacturer's tool)	2		assumption: Read from the manufacturer's life tool for clean ISO VG 68 oil.	0.1 … 50
L_req	Required life	5000	h	spec: Pump specification, minimum bearing life	1 … 1000000 h
s_0_req	Required static safety factor	1		assumption: Normal running, smooth loads.	0.5 … 4

3. Assumptions

A1. Ratings C and C_o, factors X, Y, e, X_o, Y_o and a_ISO are the manufacturer's values for the bearing, entered as reference records. (No catalogue table ships with TempestOS (§A5).)
A2. a_1 uses the ISO 281:2007 Weibull form (0.64 at 95 %, 0.25 at 99 %); ISO 281:1990 gave 0.62 and 0.21. (Edition stated on the sheet.)
A3. Loads are steady within each duty step; speed is constant. (Duty steps weighted by share of revolutions.)
A4. type (Bearing type) is taken as ball (p = 3). (Deep groove ball bearing.)
A5. F_r (Radial load) is taken as 2.5 kN. (Radial reaction from the shaft calculation.)
A6. F_a (Axial load) is taken as 0 kN. (Locating bearing elsewhere; no axial load here.)
A7. R (Reliability) is taken as 95 %. (Client reliability target.)
A8. a_iso (Life modification factor a_ISO (from the manufacturer's tool)) is taken as 2. (Read from the manufacturer's life tool for clean ISO VG 68 oil.)
A9. s_0_req (Required static safety factor) is taken as 1. (Normal running, smooth loads.)

4. Working

[18] p = 3 — ISO 281:2007 cl. 6–7
= 3
Life exponent, ball bearing

[19] P = X_1 * F_r + Y_1 * F_a — ISO 281:2007 cl. 6–7
= 1 × 2.5 kN + 0 × 0 kN
= 2.5 kN
Equivalent dynamic load (F_a/F_r ≤ e_lim)

[20] P_0 = max(X_0 * F_r + Y_0 * F_a, F_r) — ISO 76:2006 cl. 6–7
= max(0.6 × 2.5 kN + 0.5 × 0 kN, 2.5 kN)
= 2.5 kN
Equivalent static load

[21] L_10 = (C / P)^p — ISO 281:2007 cl. 6–7
= (20.3 kN / 2.5 kN)^3
= 535.39
Basic rating life, million revolutions

[22] L_10h = L_10 * 1000000 * 2 * pi / n
= 535.39 × 1000000 × 2 × π / 1500 rpm
= 5948.7 h
Basic rating life in hours

[23] a_1 = 0.95 * (ln(1 / R) / ln(1 / 0.9))^(2 / 3) + 0.05 — ISO 281:2007 cl. 9
= 0.95 × (ln(1 / 95 %) / ln(1 / 0.9))^(2 / 3) + 0.05
= 0.63791
Reliability factor (derived, ISO 281:2007 Weibull form)

[24] L_nm = a_1 * a_iso * L_10 — ISO 281:2007 cl. 9
= 0.63791 × 2 × 535.39
= 683.06
Modified rating life, million revolutions

[25] L_nmh = a_1 * a_iso * L_10h
= 0.63791 × 2 × 5948.7 h
= 7589.6 h
Modified rating life in hours

[26] s_0 = C_0 / P_0 — ISO 76:2006
= 11.2 kN / 2.5 kN
= 4.48
Static safety factor

[27] u_life = L_req / L_nmh
= 5000 h / 7589.6 h
= 0.6588
Life utilisation (required / modified life)

[28] u_s0 = s_0_req / s_0
= 1 / 4.48
= 0.22321
Static utilisation (required / actual safety factor)

5. Results

SYMBOL	VALUE	DESCRIPTION
P	2.5 kN	Equivalent dynamic load
L_10	535.39	Basic rating life (Mrev)
L_10h	5948.7 h	Basic rating life
a_1	0.63791	Reliability factor

SYMBOL	VALUE	DESCRIPTION
L_nm	683.06	Modified rating life (Mrev)
L_nmh	7589.6 h	Modified rating life
P_0	2.5 kN	Equivalent static load
s_0	4.48	Static safety factor
u_life	0.6588	Life utilisation
u_s0	0.22321	Static utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Modified rating life against required life	u_life = 0.6588	u_life ≤ 1	0.659	ISO 281:2007 cl. 9
PASS	Static safety factor against required	u_s0 = 0.22321	u_s0 ≤ 1	0.223	ISO 76:2006

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: C, C_0 (Bearing load ratings); e_lim, X, Y, X_0, Y_0 (Bearing X, Y, e factors); a_iso (Bearing a_ISO). Enter each as one so its source, edition and licence travel with it.
WARNING — input.X_1: X_1 (Radial factor X_1 for F_a/F_r ≤ e (1 for a single-row radial ball bearing)) was not entered; the template default 1 is used.
WARNING — input.Y_1: Y_1 (Axial factor Y_1 for F_a/F_r ≤ e (0 for a single-row radial ball bearing; above 0 for double-row and paired bearings)) was not entered; the template default 0 is used.

8. Verification

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Template mech.bearing-life@1.1.0 Content hash da8d1cc8ed2333eb Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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